VoiceOps 2026: ZentroMart Voice Support Agent

Capstone Technical Report & Evaluation Analysis

Consulting & Analytics Club IIT Guwahati × Gnani.ai Capstone Project

Reference Calendar Date: Fixed at `2026-10-20`

Core Technologies: Google Gemini (gemini-3.5-flash-lite & gemini-embedding-001), Gnani.ai (Prisma v2.5 STT & Timbre v2.5 TTS), Python 3.14

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1. Executive Summary

Online retail customer support in India requires handling complex, high-velocity queries across groceries and electronics: return deadlines, refund tracking, delivery rescheduling, and damaged consignments. Indian customers overwhelmingly prefer voice interactions, often speaking in Indian English or code-switching with Hindi (*Hinglish*). However, deploying standard generative LLMs directly into voice support introduces severe risks:

1. **Hallucination & Policy Traps**: Common FAQ documents often generalize refund timelines as *"5 to 7 days"*, directly contradicting binding policy documents where wallet refunds are instant within 2 hours.
2. **Calendar & Financial Math Inaccuracies**: Standard models frequently err when calculating return deadlines (calendar days) versus refund SLAs (business days, which exclude Sundays and warehouse holidays like 19 October 2026).
3. **Data Privacy & Unauthorized Disclosure**: Disclosing order information without prior identity verification violates customer trust and compliance standards.
4. **Unauthorized Commitments**: Agents frequently invent compensation (e.g. *"I have approved your refund of ■34,999"*) for high-value claims that legally require manual investigation by human priority resolution teams.

To resolve these challenges, we built **VoiceOps 2026: The ZentroMart Omnichannel Voice Support Agent**. Rather than fragmented chatbots, a single shared Python agent engine powers three production interfaces:

• **Interactive Gradio SaaS Web Application** (`app.py`) with Text Chat, Voice Assistant, Knowledge Base Browser, AGT-IDV Order Lookup, and Evaluation Lab.

• **Batch Audio CLI Runner** (`run.py` & `run.sh`) processing folders of customer voice clips into structured JSON metadata and synthesized reply WAV files.

• **Automated Benchmark Harness** (`scripts/evaluate.py`) evaluating baseline models, grounded RAG, ASR voice robustness, and multi-turn conversations against ground-truth unit tests.

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2. System Architecture & Core Modules

The architecture follows a strict separation of concerns, ensuring deterministic policy enforcement and document grounding before invoking speech synthesis:

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Customer Input Layer
(Web Chat / Microphone / Batch Audio CLI)

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Gnani Prisma v2.5 STT Engine
(Indian English & Hinglish Support)

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Speech Text Normalization
(Query Cleaning, Slang Map, Order ID Match)

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	1. AGT-IDV Caller Identity Gate	
	(State machine: phone/pincode proof)	
	2. Hybrid RRF Knowledge Retriever	
	(Dense Gemini + BM25 + Precedence)	
	3. Deterministic Python Business Engine	
	(Holiday calendar, return SLA math)	
	4. Policy Escalation & Tools Bridge	
	(Damage > 10k, Ticket Creation)	

	5. Grounded LLM Response Generator	
	(Strict document adherence)	

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Spoken Speech Normalization
(Dates -> "27 October 2026", IDs -> digits)

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Gnani Timbre v2.5 TTS Engine
(Kaveri Voice Synthesis)

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2.1 Section-Aware Knowledge Base Ingestion & Hybrid Retrieval

The knowledge base consists of 21 markdown files (`kb/*.md`) and a 70-SKU product catalog (`products.csv`). Documents are split on explicit `## [SECTION-ID]` boundaries, resulting in **154 cohesive chunks**:

- **Metadata Preservation**: Each chunk maintains document title, section ID, category, version, and document type.
- **Authoritative Precedence**: Categories are assigned strict mathematical precedence weights:

$$\text{Policy} (5) > \text{Agent Guide} (4) > \text{Help} (3) > \text{FAQ} (2) > \text{Marketing} (1)$$

- **Dense Vector Embeddings**: Chunks are embedded into 3072-dimensional vectors using `gemini-embedding-001` with persistent local disk caching (`data_index/kb_index_cache.json`).
- **Reciprocal Rank Fusion (RRF)**: Retrieval merges dense cosine similarity with sparse BM25 scores:

$$RRF(d) = \sum_{m \in \{\text{dense}, \text{sparse}\}} \frac{1}{60 + \text{rank}_m(d)} \times (1.0 + 0.15 \times \text{precedence}(d))$$

- **Hard Negative & Superseded Filtering**: Documents with `status: superseded` are strictly excluded from retrieval context, preventing outdated return policies from corrupting generation.

2.2 Deterministic Business Calendar & Order Facts

All calendar operations are handled in pure Python rather than delegating arithmetic to LLM token prediction:

- **Reference Date**: Fixed strictly at `2026-10-20`.
- **Working Days Calculation**: Mon–Sat are working days. Sundays and official warehouse holidays from `holidays.csv` (Independence Day, Gandhi Jayanti, 19 October 2026, and 09 November 2026) are skipped.
- **Return Deadlines**: Calendar days are used for return deadlines (e.g. 10 days for electronics; 15 days for standard fashion; 30 days for festive sale orders placed between 01–15 October 2026).
- **Reverse Pickup Fee**: Waived entirely for ZentroMart Plus members (`is_plus = True`) versus ■49 for regular members.

2.3 AGT-IDV Caller Identity Verification State Machine

Customer security enforces strict identity verification before disclosing personal order records:

• **Proof Challenge**: When an order number is received (Turn 2), the agent immediately challenges the caller for either the **last 4 digits of their phone number** or their **6-digit delivery pincode**.

• **Retry Enforcement**: Maximum 1 retry allowed. If 2 attempts fail, the state machine transitions to `locked_out` and terminates order inquiries without leaking order facts.

• **Privacy Redaction**: Order lookup audit logs redact verification values as `[REDACTED_SECRET]`.

2.4 Policy Escalation & Ticket Execution

The agent deterministically identifies escalation triggers:

• Consignment damage exceeding **■10,000** (e.g. 4K Smart TVs).

• Refund delays exceeding **3 business days** beyond the promised SLA.

• Repeat contacts (≥ 2 prior calls on the same order).

• Suspected account fraud or legal escalation threats.

When triggered, `create_ticket(intent="escalation")` is executed, generating a persistent ticket ID (e.g. `TKT-1002`) and routing the customer to the Priority Resolution team with a polite refusal of instant compensation.

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3. Empirical Evaluation & Benchmark Analysis

We evaluated the system using our benchmark harness (`scripts/evaluate.py`), testing across four distinct configurations:

1. **Step 1 Baseline**: Direct LLM generation using general knowledge without knowledge base documents or tool access.
2. **Step 2 Grounded RAG (Clean Text)**: Augmented with hybrid retrieval over all 154 indexed chunks.
3. **Step 3 Grounded RAG (ASR Robustness)**: Evaluated on simulated speech-to-text transcripts (`asr_text`) featuring transcription disfluencies and number formatting.
4. **Step 4 Multi-turn Conversations**: Full conversational agent tested across multi-turn sessions with identity verification, order lookup, and escalation triggers.

3.1 Consolidated Benchmark Results

Evaluation Metric	Step 1: Baseline	Step 2: Grounded RAG	Step 2: RAG (ASR Robust)	Step 3: Multi-turn Convs
Sample Size	5 Questions	5 Questions	5 Questions	22 Turns (5 Convs)
Must-Include Accuracy	**20.0%**	**80.0%**	**80.0%**	**100.0%**
Must-Not-Include Accuracy	100.0%	100.0%	100.0%	100.0%
Decision Accuracy	100.0%	100.0%	100.0%	**95.45%**
Tool Call Accuracy	N/A	N/A	N/A	**100.0%**
Overall Pass Rate	**20.0%**	**80.0%**	**80.0%**	**95.45%**
Avg Turn Latency	919.4 ms	2,098.3 ms	3,470.2 ms	3,162.6 ms

3.2 Key Findings & Comparative Insights

1. **Failure of the Baseline Without Documents**:

The baseline model scored only **20.0%** on must-include accuracy. On question `PQ0003` (*"How fast are refunds to the ZentroMart Wallet?"*), the baseline model failed to state the 2-hour SLA. On product catalog queries (`PQ0002`, `PQ0005`), the baseline lacked

product specifications and warranty durations.

2. **Precedence Defense & The FAQ Trap**:

In Step 2 Grounded RAG, hybrid precedence boosting prioritized policy `REF-2` over `FAQ-1`. The agent correctly quoted:

> *"Refunds to your ZentroMart Wallet are instant, processed within 2 hours of initiation."*

It achieved a **0.0% hard-negative hit rate**, completely rejecting the misleading 5-to-7 day FAQ timeline.

3. **Robustness to ASR Speech Noise**:

Testing on `asr_text` (e.g. *"how fast are refunds to the zentromart wallet"*, *"so sku ele-zero zero one to pe zentromart assured hai kya"*) achieved identical **80.0%** rule satisfaction, proving that our query slang normalization and fuzzy SKU matcher effectively shield the agent from speech recognition artifacts.

4. **Stateful Conversation Precision**:

Across 22 turns in Step 3, the agent achieved **100.0% must-include accuracy** and **100.0% tool call accuracy**. Identity verification was enforced without exception on Turn 2, proof was validated on Turn 3, and high-value damage claims were escalated to Priority Resolution with ticket generation on Turn 4.

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4. Engineering Trade-Offs & Production Decisions

1. **Free-Tier API Rate Limiting & Resilient Multi-Model Fallback**:

Google Gemini free-tier endpoints impose strict rate limits (20 requests/day on `gemini-2.5-flash`). To guarantee 100% service uptime during live demonstrations, our `GeminiProvider` implements intelligent backoff: if a model returns a daily quota exhaustion delay ($>60\text{ s}$), it immediately falls back to `gemini-3.5-flash-lite`, ensuring seamless continuation without throwing runtime exceptions.

2. **Explicit Voice Engine Enforcement**:

Per project rules, all speech processing strictly invokes Gnani.ai (`gnani-prisma-v2.5` for STT and `timbre-v2.5` with voice `Kaveri` for TTS). Explicit model and voice parameters are passed to every REST call.

3. **Empty Audio Recovery**:

When background noise or silent audio is received, the agent generates a friendly clarification reply (*"Sorry, I didn't catch that. Could you please say it again?"*) and synthesizes it into audio, avoiding pipeline crashes.

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5. Conclusion

VoiceOps 2026 demonstrates that production-grade voice customer support requires much more than an off-the-shelf LLM. By fusing **authoritative document precedence**, **deterministic business calendar arithmetic**, **strict caller identity verification**, and **high-fidelity Gnani speech synthesis**, the ZentroMart Voice Support Agent achieves an exceptional **95.45% overall pass rate** on multi-turn conversations while delivering a natural, low-latency conversational experience.